

NETWORK TOPOLOGY

Network Topology

- Geographical representation of the links.
- Topology is physical **layout** of **computers, cables and other connected devices** on a network.
- Types
 1. Physical Topology
 2. Logical Topology

Types of Topology

- **Physical Topology:**
 - **actual layout** of the computer cables and other network devices.
- **Logical Topology:**
 - the way in which the **network appears** to the devices that use it.
 - Refers to **how data is actually transferred** in a network.

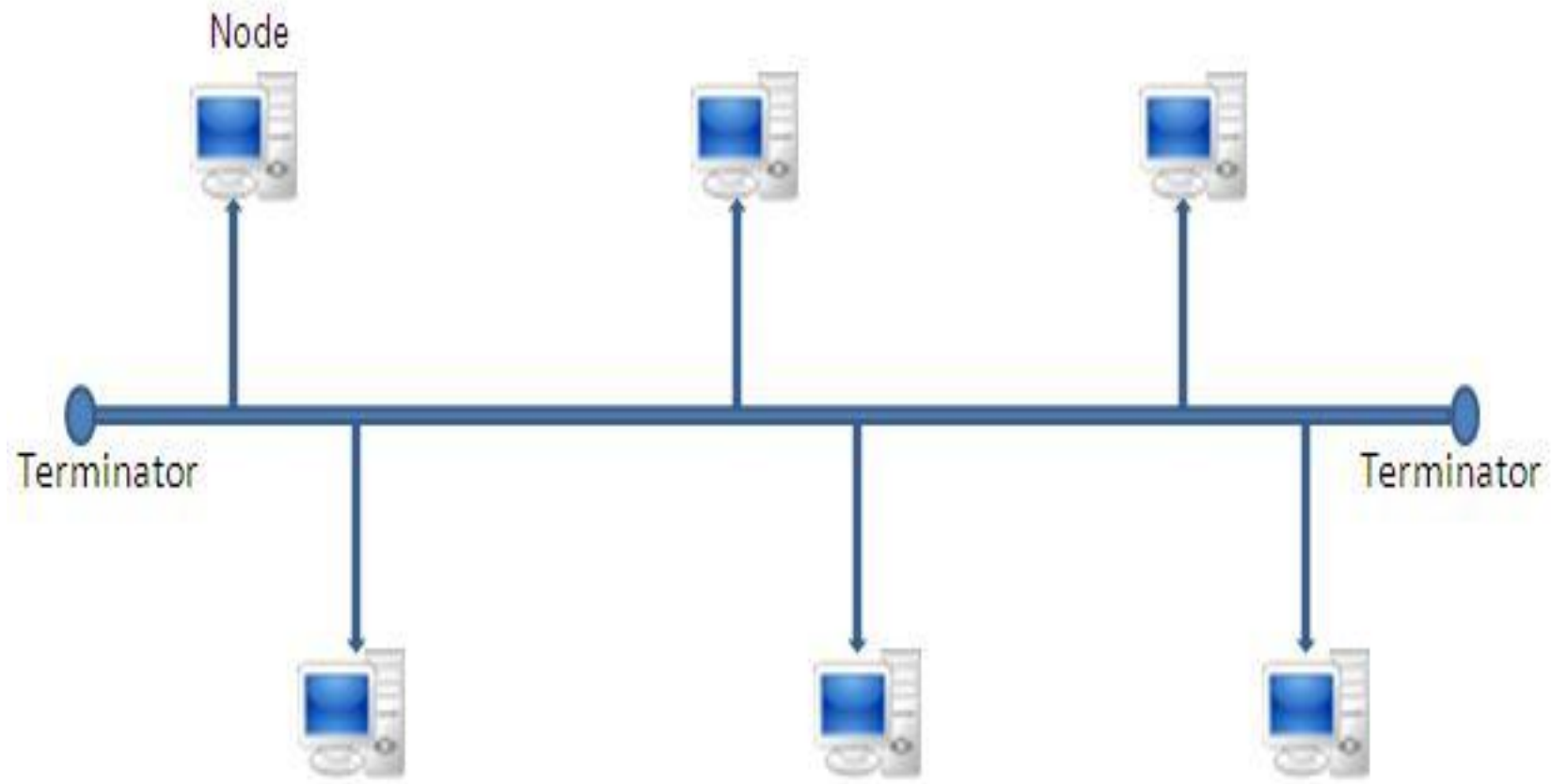
Selection Criteria

- **Size (no of node) of the system.**
- **Cost of the components and service required.**
- **Management of network.**
- **Architecture of network.**
- **Cable type.**
- **Expandability of the network.**
- **The desired performance.**
- **Reliability.**

Types of Network Topology

- Here, some logical layout of topology.
 - **Bus**
 - **Ring**
 - **Star**
 - **Mesh**
 - **Tree and Hybrid**

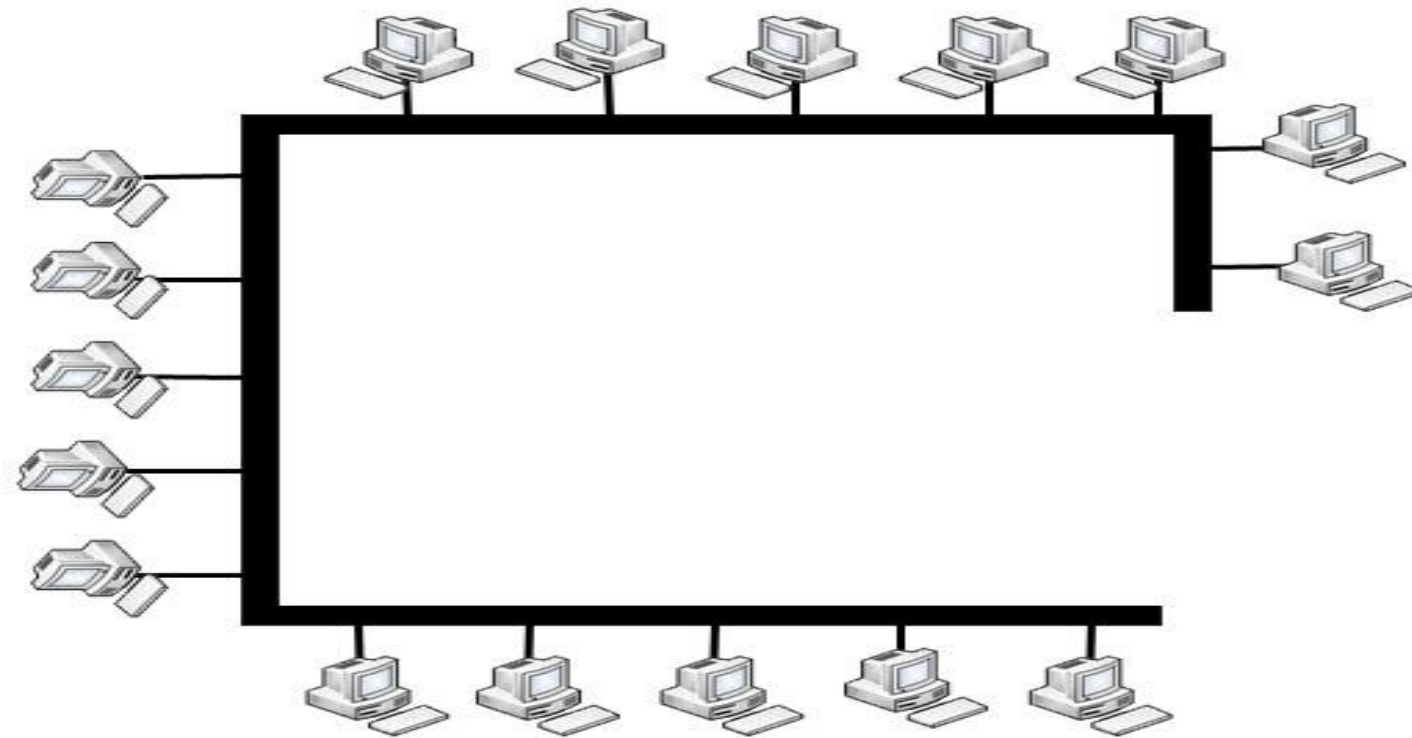
Bus Topology



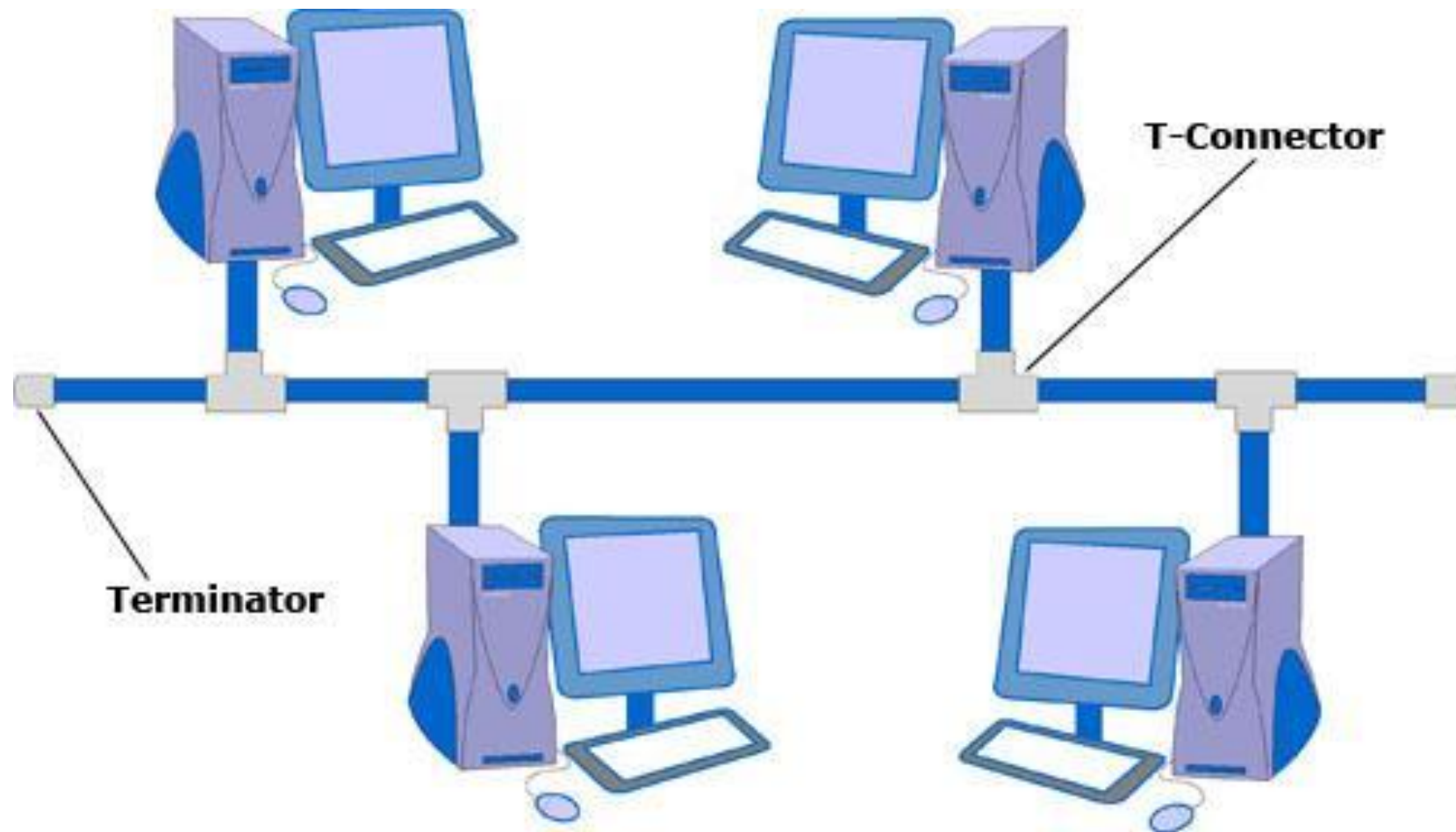
Bus Topology

- Also called as **Linear topology**.
- A bus topology is **multipoint**.
- Here one long cable act as a **backbone** to link all the devices.
- Devices are connected to the backbone by **drop lines** and **taps**.
- **Drop line**- is the connection b/w the devices and the cable.
- **Tap**- is the splitter that cut the main link.
- This allows **only one device to transmit at a time**.

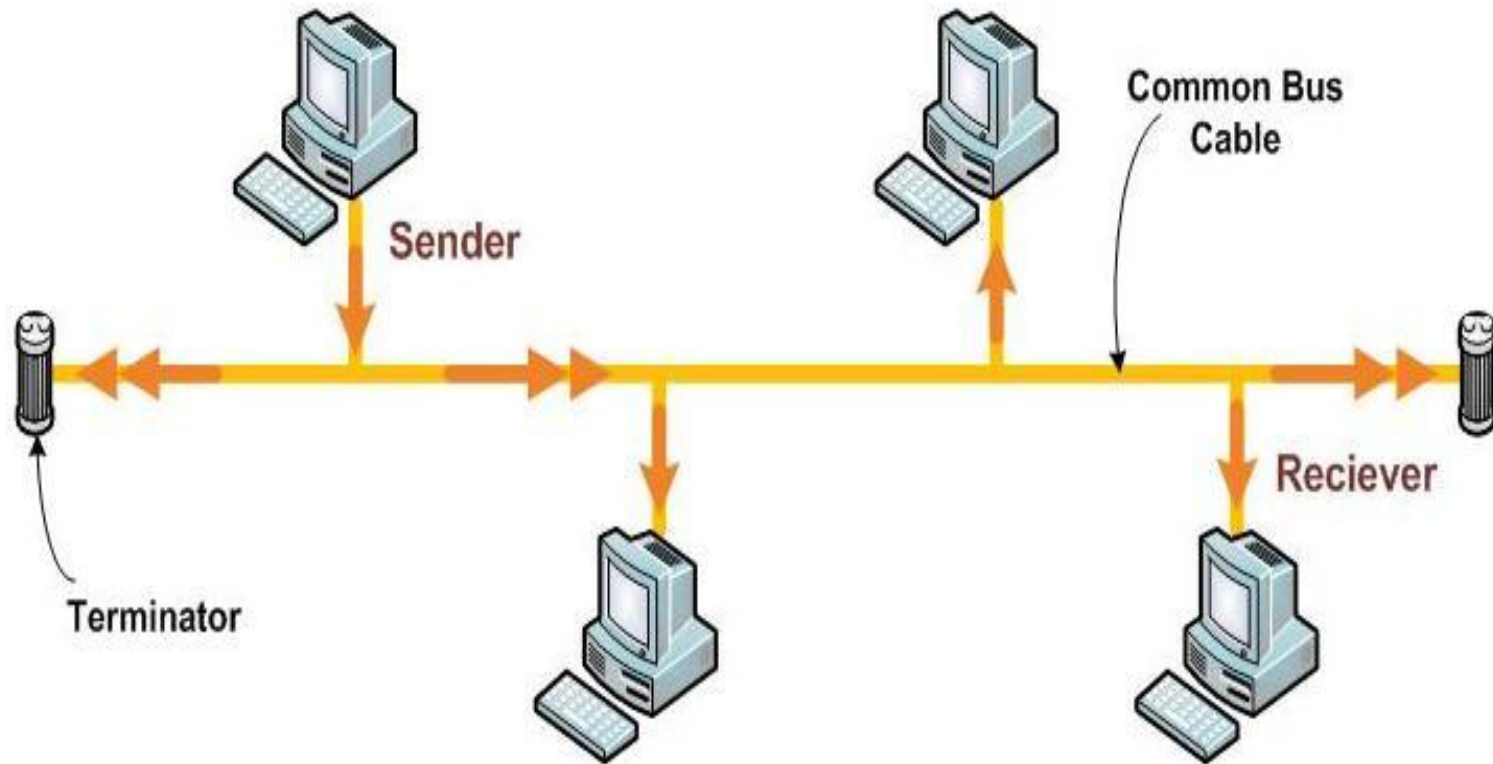
Bus Topology



Bus Topology



Bus Topology



Bus Topology

- If a device want to communicate with other device on the n/w, sends a **broadcast message** onto the wire all other devices receive the message.
 - But only the intended devices accepts and process the message.
 - Bus is **passive** topology.
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- [Bus Topology](#)
 - [Bus Failure](#)

Bus Topology

• *Advantages:*

1. Simple, reliable for small networks
2. Ease of installation
3. Less cabling, less expensive.
4. Suitable for temporary and small networks.

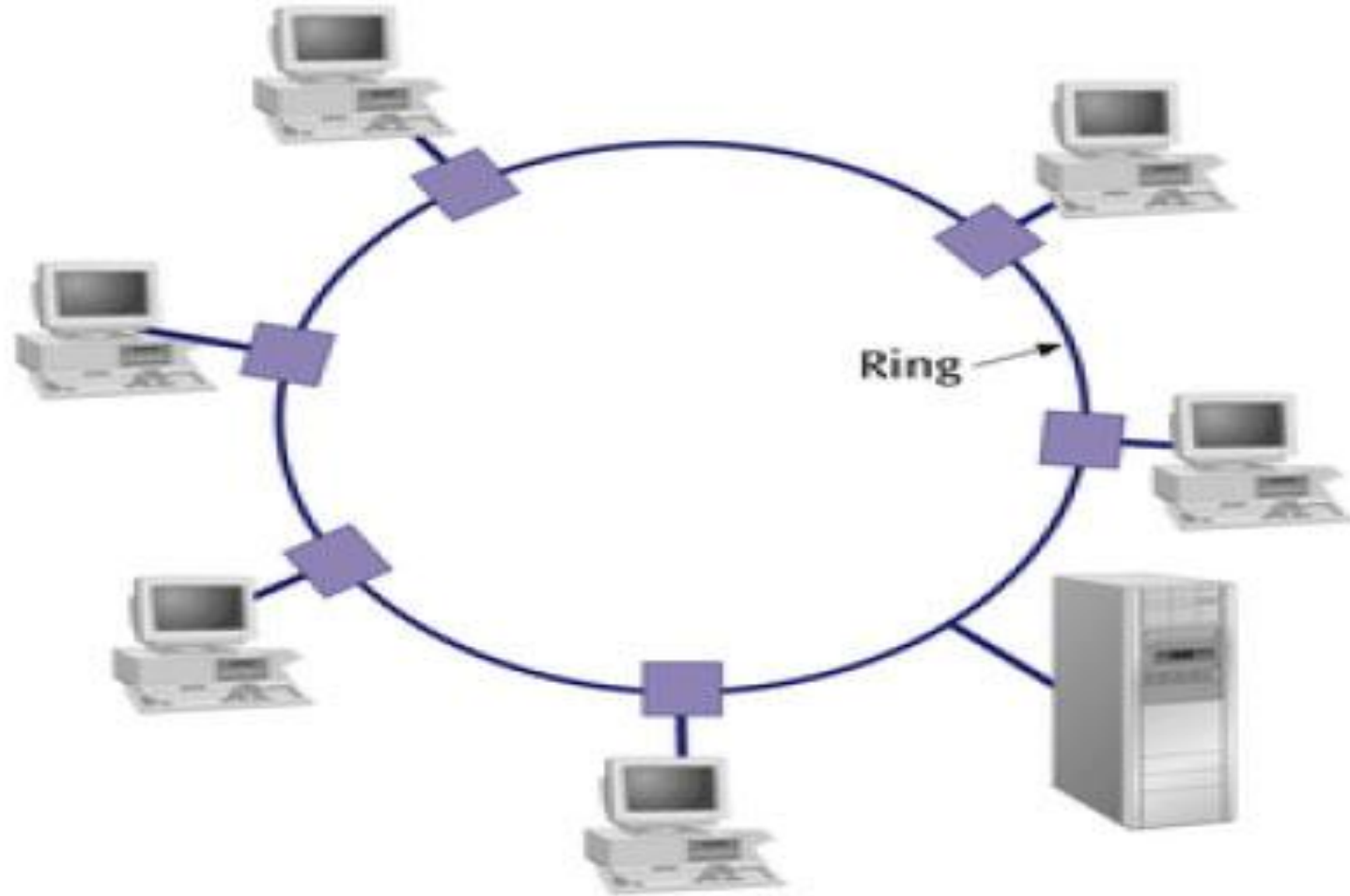
Disadvantages:

1. Difficult reconfiguration and fault isolation.
2. Difficult to add new devices.
3. Signal reflection at tap can degradation in quality.
4. If any fault in backbone can stops all transmission.
5. Only one computer can transmit at a time.
6. Limited cable length and number of stations.

Bus Topology

- *Application:*
- Most computer motherboard.

Ring Topology



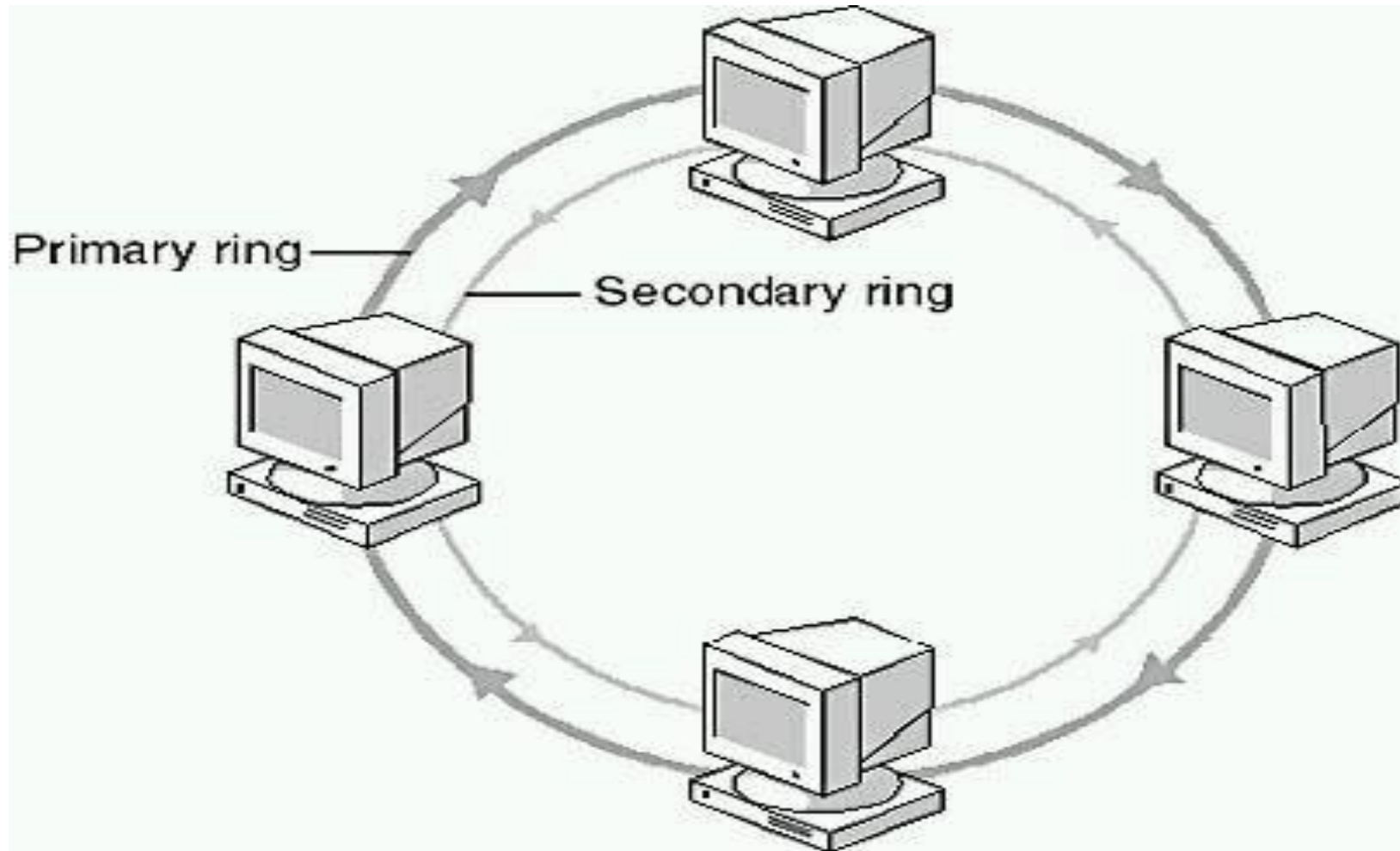
Ring Topology

- In Ring topology each node is connected to the two nearest nodes so the **entire network forms a loop**.
- The signal is passed in **one direction** from one device to another device until it reaches the destination.
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- Ring is the **Active Network**, because each computer retransmits what it receives from previous computer.
- To add or delete a device requires changing only two connections.

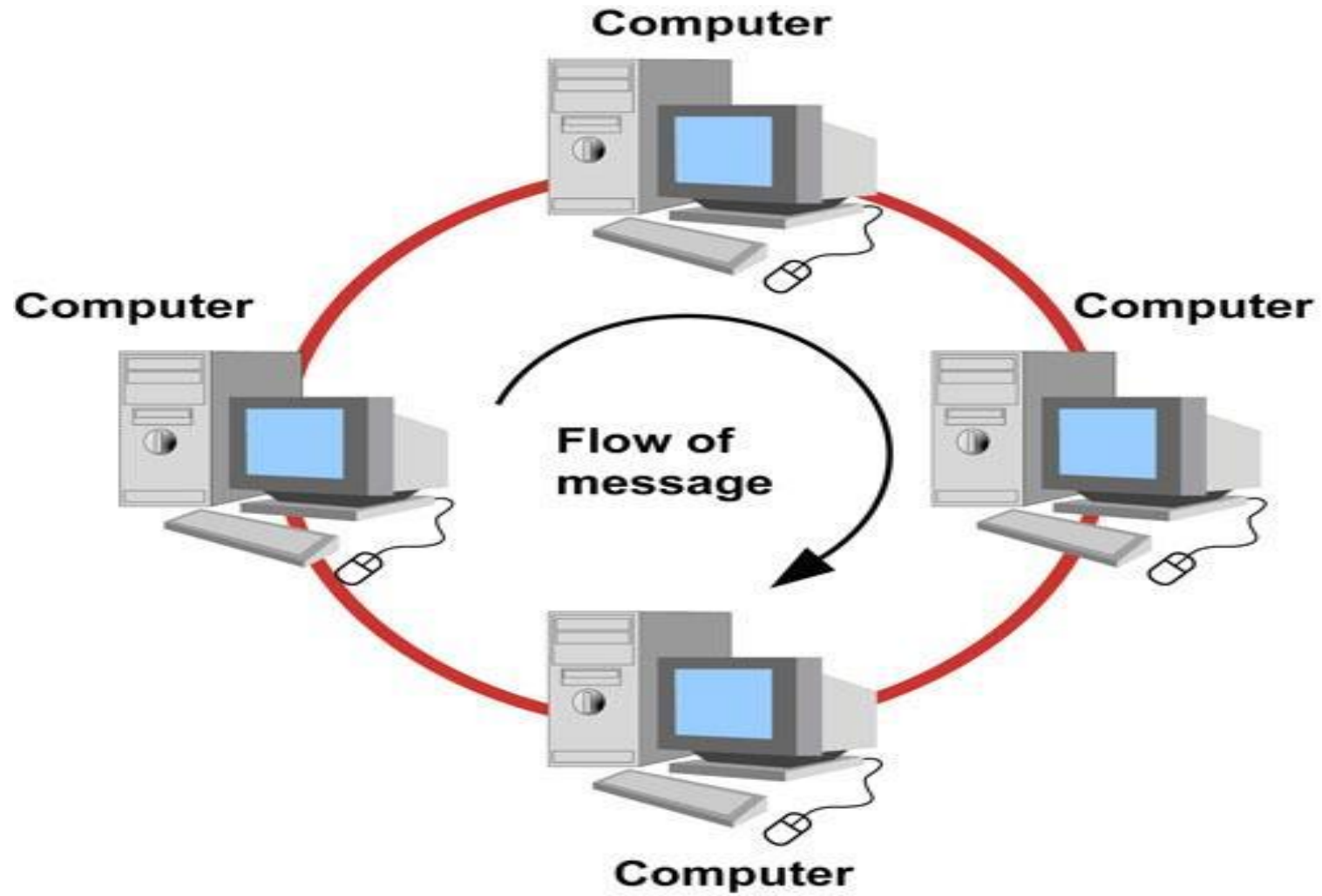
Ring Topology

Token ring
Failures

Ring Topology



Ring Topology



Token Ring Topology

- Variation of Ring Topology.
- Token Ring

Ring Topology

- *Advantages :*

1. Time to send data is known
2. No data collisions
3. Easy to install.
4. Easy to reconfigure.
5. Fault identification is easy.

- *Disadvantages :*

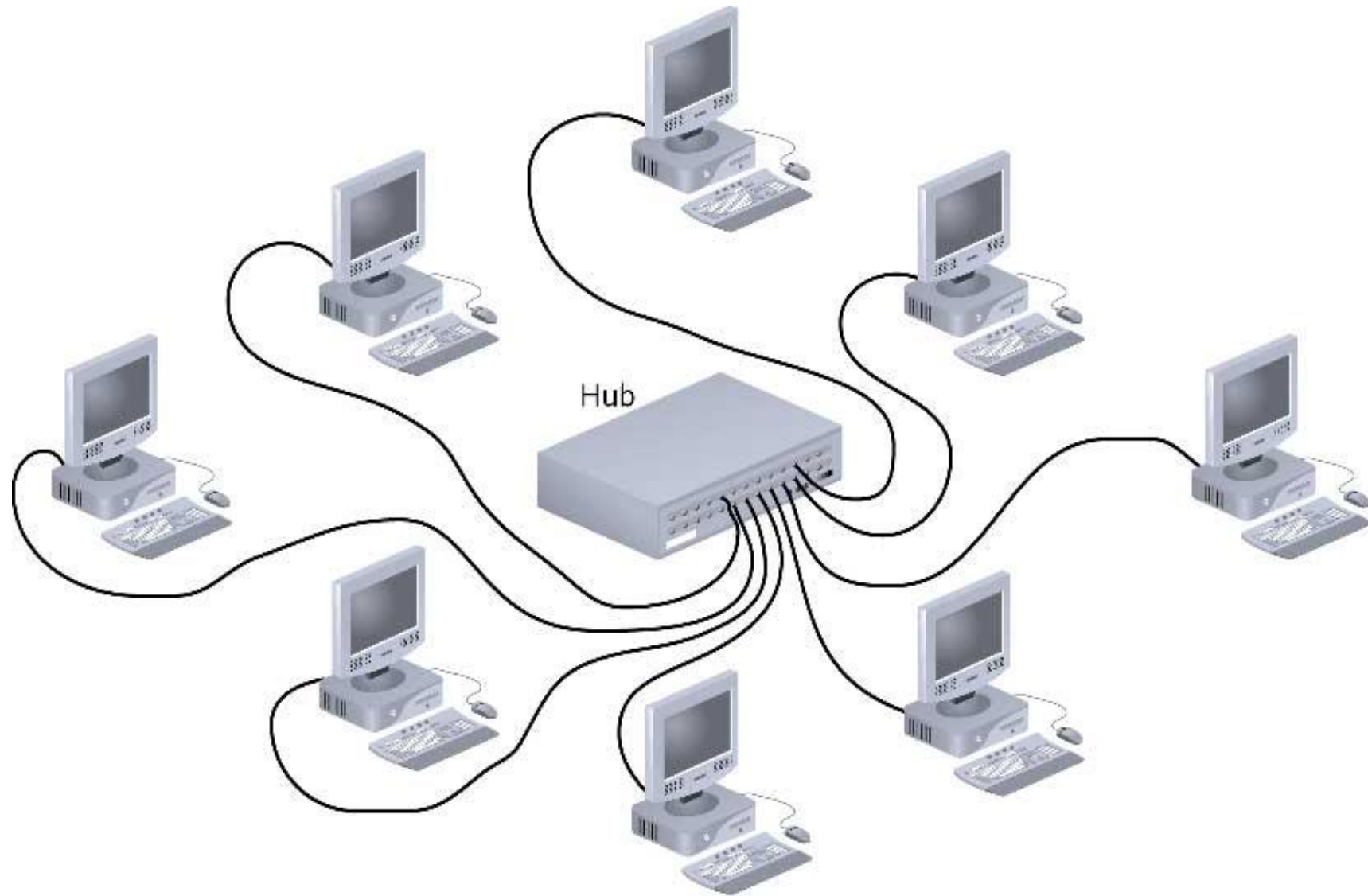
1. Unidirectional traffic.
2. Break in a single ring can break entire network.
3. Expensive.
4. Requires more cable and network equipment at the start

Ring Topology

- **Applications:**

- Ring topologies are found in some office buildings or school campuses.
- Today high speed LANs made this topology **less popular.**

Star Topology

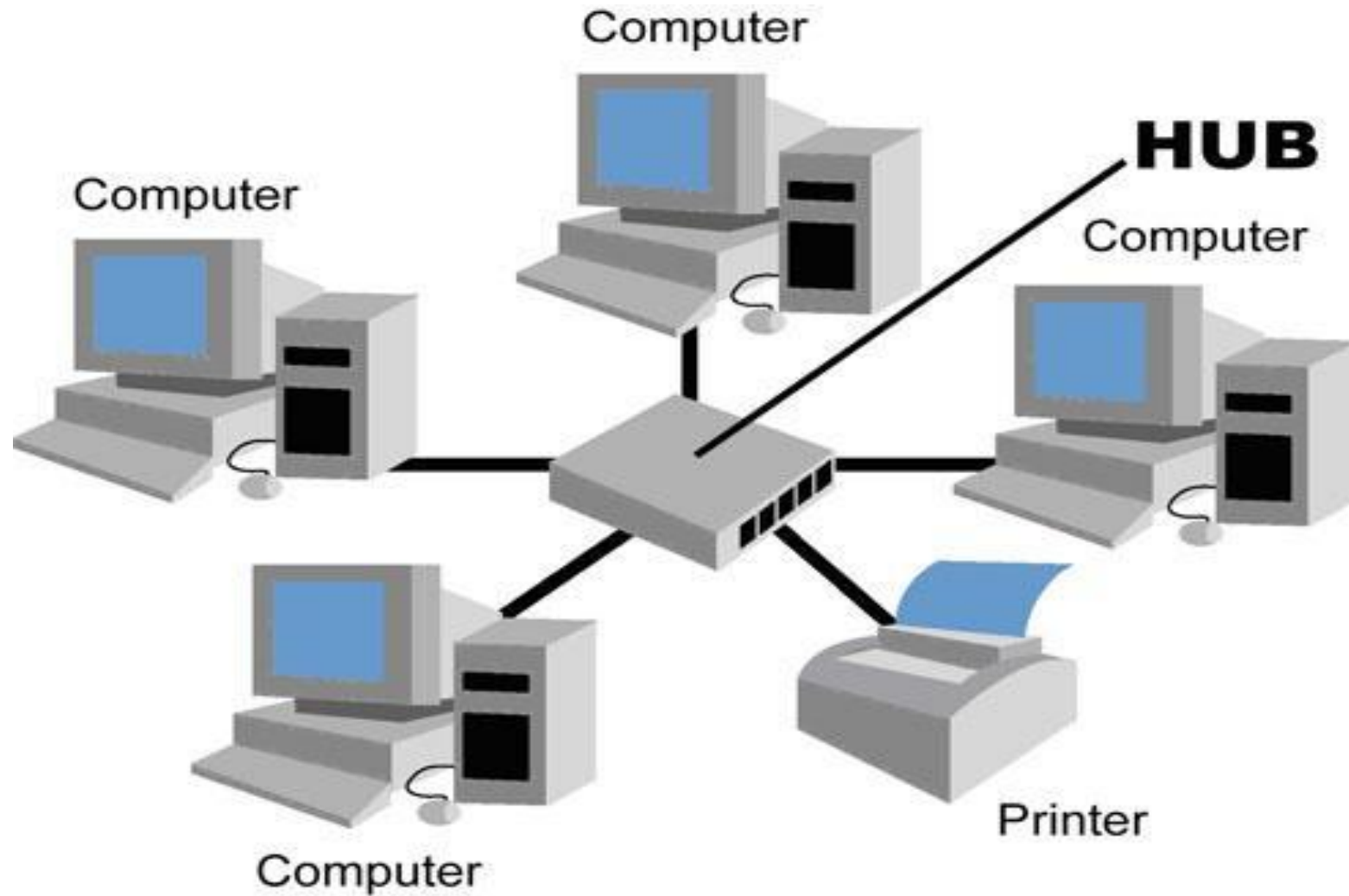


Star Topology

- Here each device has a dedicated point-to-point link to the central controller called **“Hub/Switch”(Act as a Exchange)**.
- **There is no direct traffic between devices.**
- The transmission occurs only through the central **“hub”**.
- When device one wants to send data to device 2; First sends the data to hub. Which then relays the data to the other connected device.

Star
Topology

Star Topology



Star Topology

- *Advantages :*

1. Good option for modern networks.
2. Low setup costs.
3. Installation, management and configuration is easy.
4. Less cabling is needed.
5. **Robustness.**(if one link fails, only that links is affected. All other links remain active)
6. Easy to **fault identification** & to remove parts.
7. **Scalable:** No disruptions to the network when connecting(or) removing devices.

Star Topology

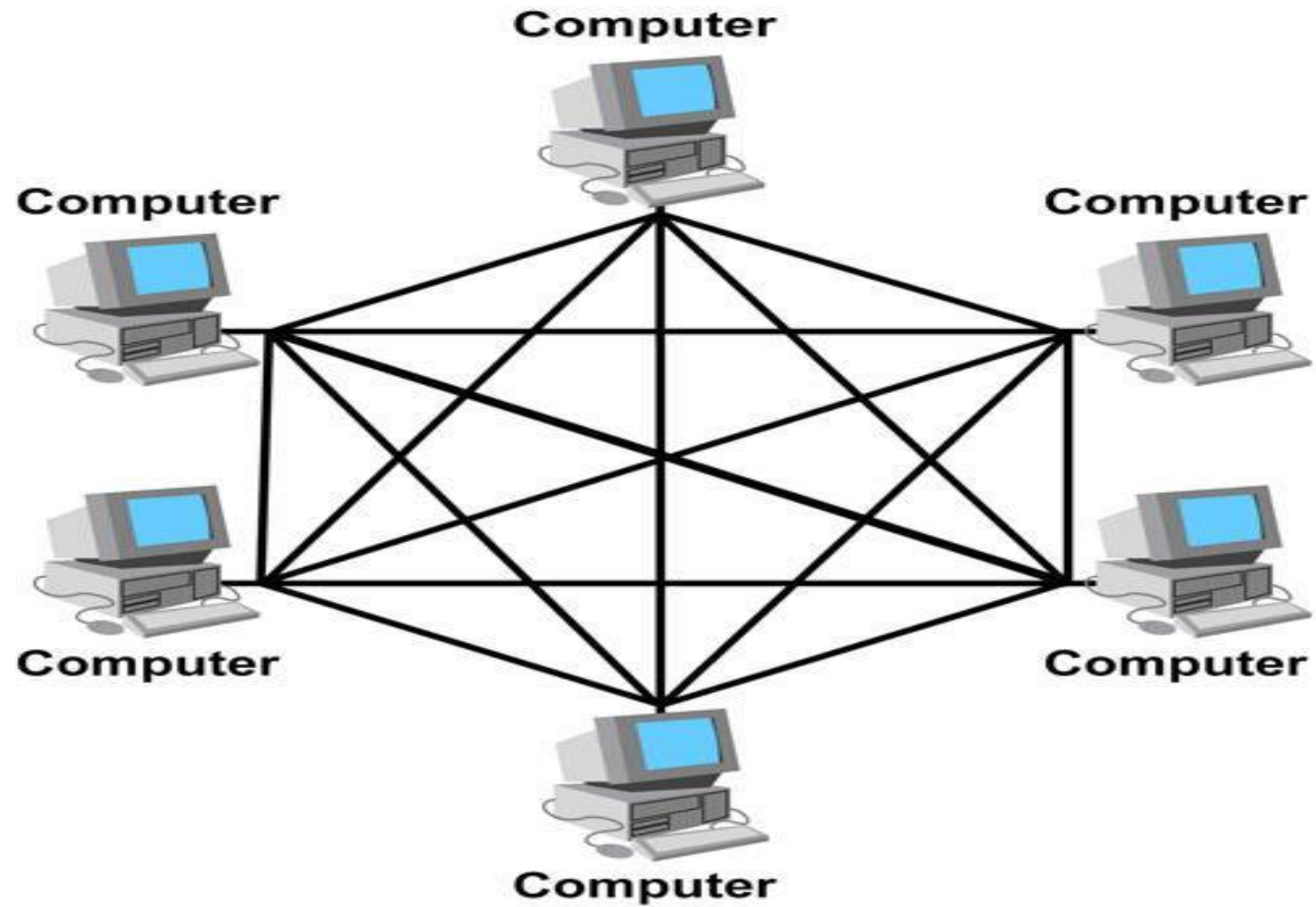
- *Disadvantages :*

1. Requires more cable than the bus
2. **Dependency:** Hub is a single point of failure. When it goes down. The whole system is dead.

Applications

- Star topology used in Local Area Networks (LANs).
- High speed LAN often use STAR.

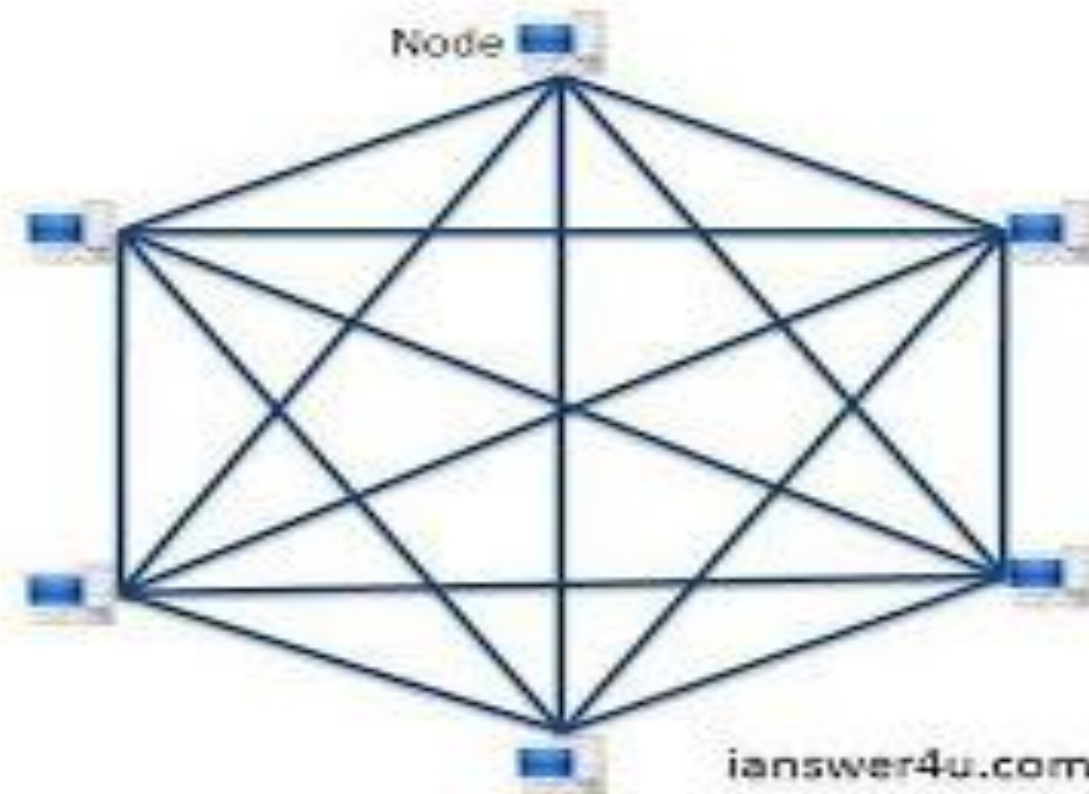
Mesh Topology



Mesh Topology

- Here every device has a **point to point** link to every other device.
- Node-1 must be connected with **$n-1$** nodes.
- A fully connected mesh can have **$n(n-1)/2$** physical channels to link **n** devices.
- It must have **$n-1$** i/o ports.

Mesh Topology



Mesh Topology

Advantages:

1. They use dedicated links so each link can only carry its own data load. So **traffic problem** can be avoided.
2. **It is robust**. If **any one link get damaged** it cannot affect others.
3. It gives privacy and security.(Message travels along a dedicated link)
4. Fault identification and fault isolation are easy.

Mesh Topology

- *Disadvantages:*

1. **Cabling cost is high:** The **cabling** and **I/O ports** required are more, since every device is connected to each devices through dedicated link.
2. **Wiring is very complicated**
3. Hardware required to connected each device is highly expensive.
4. Troubleshooting a failed cable is tricky

Mesh Topology

- *Applications:*

1. **Telephone Regional office.**
2. **WAN.(Wide Area Network).**

Tree Topology

- Alternatively referred to as a **star bus** topology.
- Tree topology is one of the most common network setups that is similar to a bus topology and a star topology.
- A tree topology connects multiple star networks to other star networks.

- **Hierarchal Model**

- Advantages

- Scalable
- Easy Implementation
- Easy Troubleshooting

Tree Topology

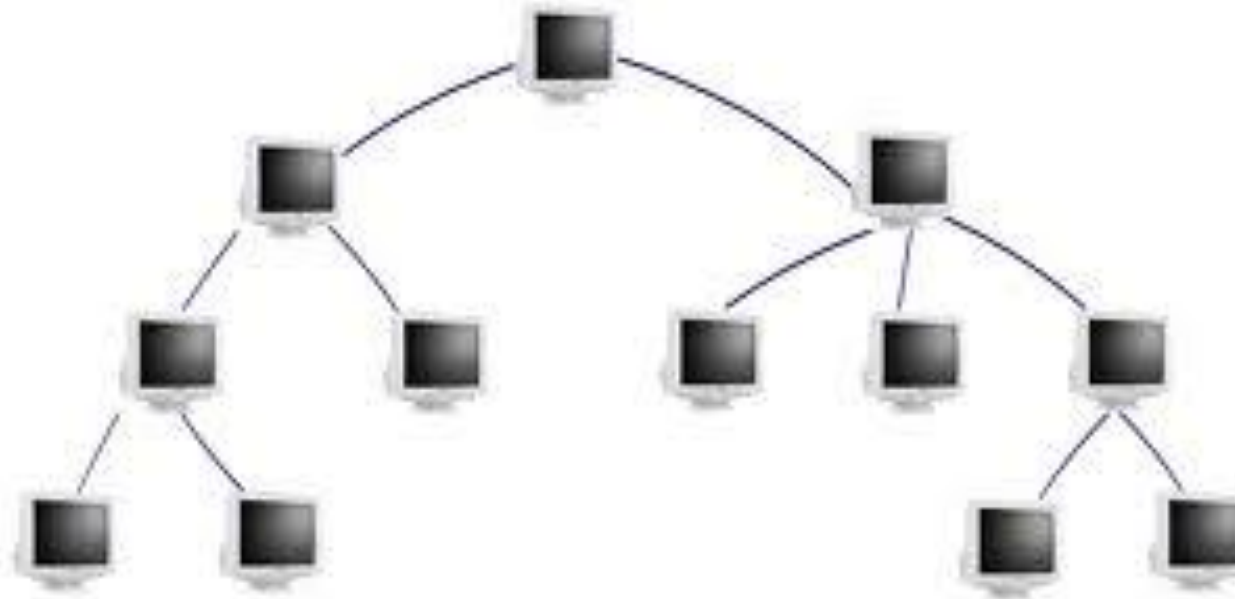


Diagram - Tree Topology

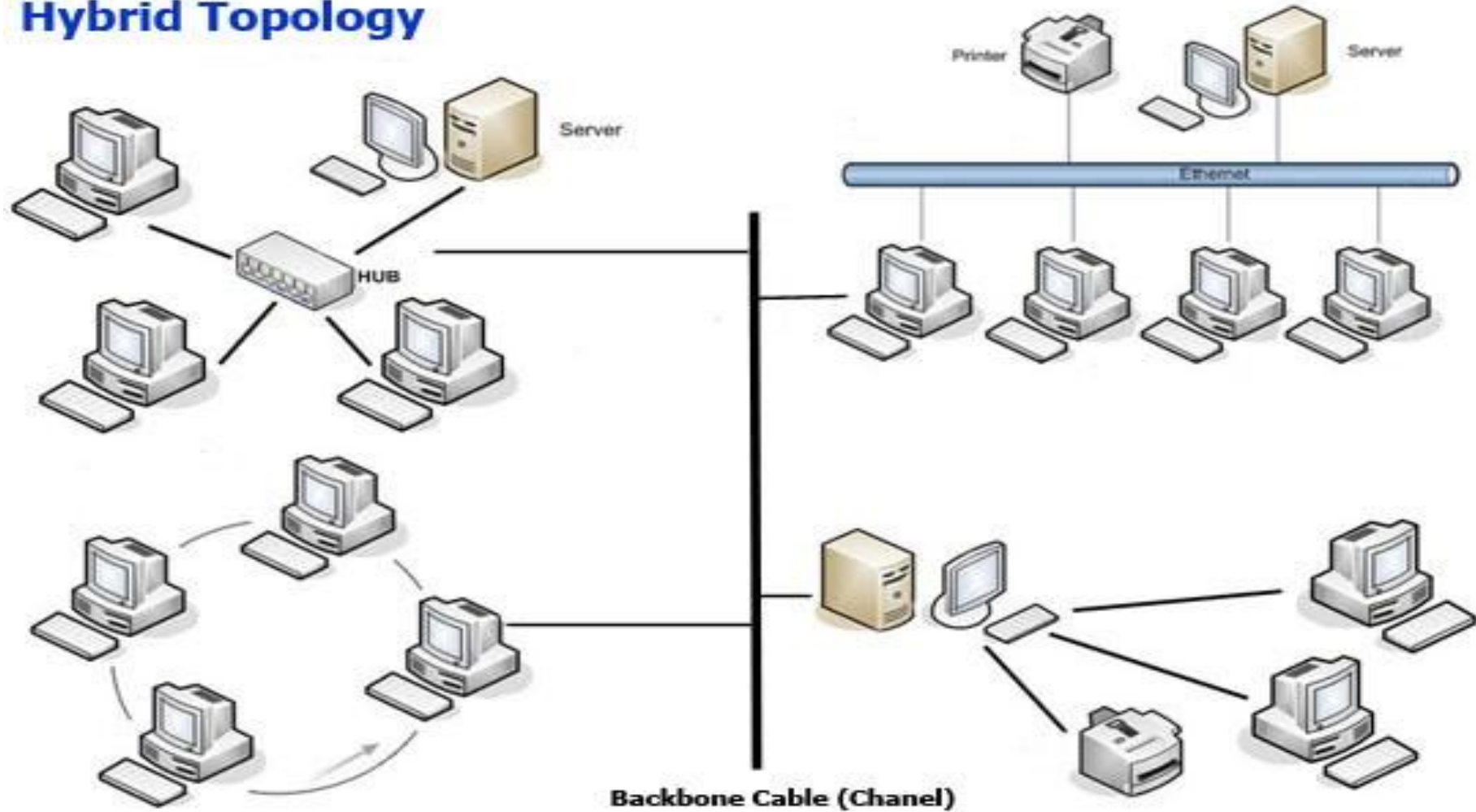
Hybrid Topology

- A network which contain all type of physical structure and connected under a single backbone channel.

Hybrid Topology

- a

Hybrid Topology



Question

- Draw **star-bus** topology connecting **three star networks having three computer into two stars and two computers in one star?**

Draw a neat labeled sketch of a **hybrid topology** connecting **one star network of 4 computers**, **one ring network of 5 computers** and **one bus network of 5 computers**.

Considerations for choosing topology

- **Money**-Bus n/w may be the least expensive way to install a n/w.
- **Length**-of cable needed- the linear bus n/w uses shorter lengths of cable.
- **Future growth**-with star topology, expanding a n/w is easily done by adding another devices.
- **Cable type**-most common used cable in commercial organization is twisted pair. Which often used with star topologies.

- Full **mesh topology** is theoretically the best since every device is connected to every other device.(thus maximizing speed and security. however, it quite expensive to install)
- Next best would be **tree topology**, which is basically a connection of star and bus.

Network Topology

